CHARLOTTE HOUDAILLE

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SUMMARY

Final-year engineering student in Creative Technology (ESILV / IFT), combining mechanical design, rapid prototyping, electronics/IoT, artificial intelligence and biomaterials. Seeking a 6-month end-of-studies internship in R&D and innovation starting February 2027.

PROFESSIONAL EXPERIENCE

Engineering Intern – Territorial Resilience & Climate **April – August 2026**
Ville du Lamentin / MyGwave *Martinique, France*

- **IoT & rapid prototyping:** 3D design and field deployment of an urban environmental sensor network.
- **Autonomous hardware:** prototyping of a solar-powered weather station and autonomous solar retrofits for tsunami evacuation signage.
- **Data & web:** data cleaning, processing and visualization in Python, with an online monitoring web portal.

R&D Engineering Intern – AMR Project (Autonomous Mobile Robot) **June – July 2024**
Fives Conveying *Montévrain, France*

- **3D modeling & mechanical design:** SolidWorks modeling, mechanical calculations and AMR chassis optimization.
- **3D animation & prototyping:** 3D design and animation in Blender for robot demonstration; participation in functional prototyping.

EDUCATION

Diplôme d'Ingénieur (Master of Engineering) & MSc in Creative Technology (Elective: AI) **2022 – 2027**
ESILV, at IFT (Institute for Future Technologies) *Paris La Défense, France*

- **Curriculum:** integrated preparatory program (2 years), followed by the engineering cycle.
- **Soft skills:** 4 development programs, including a certification in Design Thinking.

Erasmus Exchange Semester – Management & Engineering **Sept. 2024 – Jan. 2025**
VUM (Varna University of Management) *Varna, Bulgaria*

PROJECTS

- **3D Bioprinting (COSMYX Nova):** R&D and mechanical retrofit of a conventional 3D printer for living cell-laden bio-ink extrusion; design and integration of a dedicated extruder.
- **Biomaterials (Mycelium):** applied research on mycelium cultivation for the development of alternative bio-leather and structural applications in space habitats.
- **Product Development & Crowdfunding (ORIKA Project):** end-to-end design of an origami leather wallet: physical prototyping, industrialization, supply chain logistics and Kickstarter campaign.

SKILLS

- **Hard skills:** CAD / 3D (SolidWorks, Fusion 360, Rhino 3D, Blender, BambuStudio); electronics & IoT (KiCad, ESP32, microcontrollers, sensors); programming (Python, C#, SQL, CSS); Machine Learning & Data (Pandas, PyTorch, Scikit-Learn).
- **Soft skills:** rapid prototyping / maker mindset, project management, curiosity & adaptability, teamwork.

LANGUAGES

- **French:** native speaker.
- **English:** TOEFL ITP level C1 (10 years lived in Australia).
- **Spanish:** A2 level.

LEADERSHIP & EXTRACURRICULAR ACTIVITIES

- **ESILV Student Union (BDE):** Event & Travel Committee (2023 – 2026).
- **LéoSurvival (Sports Association):** President (2023 – 24) then Treasurer (2025 – 26); overall organization of the 'Raid du Loup Perdu' adventure race (350 participants, safety protocols, 6 to 22 km routes).

INTERESTS

- **Outdoor sports:** swimming, hiking / trekking, Coastal Boating License.
- **Travel & culture**